Comparison of Different Recommendation Models based on Amazon Platform Dataset

Qianyi Wang
Chinese University of Hong Kong (Shen Zhen)
Shenzhen, Guangdong, China
121040071@link.cuhk.edu.cn

Abstract

As the society enters the era of" information explosion", the problem of information overload has become more and more prominent, and different recommendation systems are born. However, it is particularly important to choose the right recommendation model because different recommendation models have different accuracy for binary classification problems. This paper mainly compares the Amazon platform dataset by using logistic regression, decision tree, random forest, and deep crossing models and their mixed models, and finally measuring their performance. The results show that the combination model by using stacking method with a deep crossing weight of 0.70, logistic regression weight of 0.10, and random forest weight of 0.20 achieved the highest accuracy of 0.7567.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches.

Keywords

recommendation system, logistic regression, decision tree, random forest, deep crossing, hybrid model

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1 Introduction

In the digital age, the exponential growth of information has given rise to significant challenges related to information overload. To address this issue, recommendation systems have become essential in filtering and personalizing content for users. E-commerce platforms like Amazon rely on recommendation algorithms heavily to enable personalized recommendations and provide users with relevant suggestions.

This study aims to systematically compare the performance of various recommendation models on the Amazon platform dataset, focusing on logistic regression, decision tree, random forest, and deep crossing models, as well as their hybrid combinations. We will examine the models' ability to accurately predict user preferences and recommend relevant items, emphasizing factors such as accuracy, scalability, and computational efficiency. By selectively training key features from the dataset, this comparison will offer valuable insights into which models are best suited for real-world e-commerce applications.

This literature review aims to synthesize the existing research on recommendation systems, highlighting the progress made in various approaches, identifying persistent gaps, and suggesting potential avenues for future exploration.

Latha and Rao (2023) developed an automated product recommendation system using a modified Convolutional Neural Network (CNN) tailored for e-commerce. The modified CNN model achieves a mean accuracy of 97.40% on the Amazon product review dataset [1]. Also, Wang et al. (2016) introduces a model that leverages LR and GBDT algorithms to effectively process and extract features from the noisy mobile data. Their approach involves novel data preprocessing techniques tailored for mobile scenarios [2]. Moreover, Wu (2021) investigates various models, including the Multi-Level Regression (MLR) model, Deep Interest Network (DIN) model, Deep Interest Evolution Network (DIEN) model, and Deep Sequential Interest Network (DSIN) model. The analysis reveals that each model's attention mechanism improvements, driven by insights into user behavior, have led to increased AUC (Area Under the Curve) and substantial economic benefits [3]. Besides, Hu (2017) demonstrates that the Bi-generator neural network can effectively learn latent feature vectors and offers performance comparable to recent neural network-based proposals. The study applied this approach to the MovieLens dataset, the results suggest that the Bi-generator model holds promise for further development and application [4]. Finally, Yun et al. (2018) proposes a method that leverages user review data extracted through opinion mining for product recommendation systems and evaluate its effectiveness by testing product recommendations on Amazon data [5].

2 Method

Figure 1 titled "Experimental Procedure" illustrates the process flow of a machine learning experiment. Initially, data preprocessing is performed, followed by the division of the dataset into training, validation, and testing sets. Four different models—logistic regression, decision tree, random forest, and deep crossing—are run independently on the prepared data. Moreover, a hybrid model is constructed by using the outputs of these individual models. Finally, the results from all models are compared to evaluate performance and draw conclusions.

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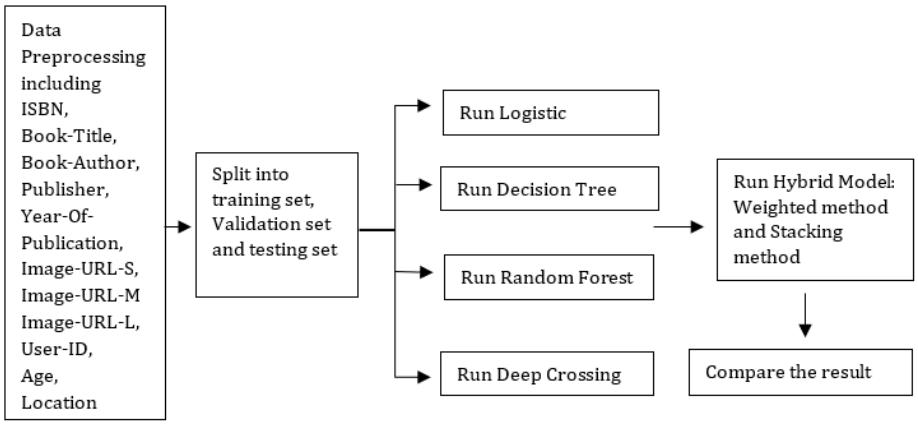

Figure 1: Experimental procedure.

Table 1: Introduction for each table and key variables.

Table name	Description	Key variables
Users table	Describes the information of users of Amazon website. The total number of users is 2140.	User-ID (User unique identifier) Age (Users' ages) Location (Users' location)
Books table	Describes the basic information of books. The total number of books is 623, and 8 features in total.	ISBN (Books' unique identifier) Book-Title (Name of each book) Book-Author (Author of each book) Publisher (The publisher's name that published the book. Year-Of-Publication (The year that publisher published the book) Image-URL-S (The URL link of the small size of book cover) Image-URL-M (The URL link of the medium size of book cover) Image-URL-L (The URL link of the large size of book cover)
Rates table	Describe how each user rates the book. The total number is 1341780, and nearly 95% of the rates are missing.	Book-Rating (Each user's rating of the book) User-ID (User unique identifier) ISBN (Book unique identifier)

2.1 Data description

Table 1 titled "introduction for each table and key variables" illustrates the information of three CSV files. The first file called "Users", which contains anonymized user information, including unique user IDs, locations, and ages. The second file called "Books", which provides detailed information about the books, such as the ISBN, book title, author, publisher, year of publication, and associated book images. Lastly, the third file contains user ratings, where each user assigns a score ranging from 0 to 10 to the books that they have interacted with. The ratings file links the anonymized user IDs from the "Users" file to the ISBNs from the "Books" file. It also provided a structured way to analyze user preferences across the dataset.

2.2 Data preprocessing

To prepare the dataset for analysis, we first read the three files and merged them based on User-ID and ISBN. Subsequently, we removed the book URL as it was deemed non-informative for the model. All missing values were eliminated to ensure data quality and minimize noise, which is crucial for improving the model's accuracy.

For the remaining features, we performed the following preprocessing steps:

ISBN, Image-URL-S, Image-URL-M and Image- URL-L: These features are discarded as it is not needed for the analysis.

User-ID and Publisher: These categorical variables were one-hot encoded to convert them into numerical format.

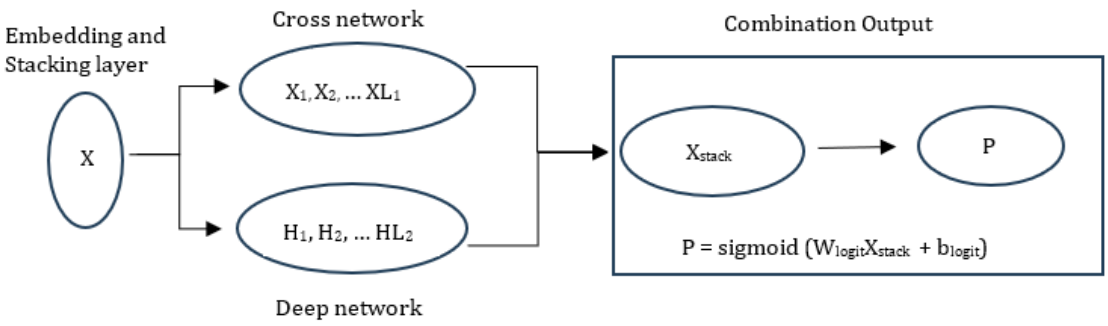

Figure 2: Deep crossing model structure.

Age: We used quartile-based methods to handle outliers by capping ages that fall outside the interquartile range to the nearest quartile limits. The age data was then normalized using Min-Max scaling.

Location: This feature was divided into City, State, and Country, and each component was encoded using label encoding. The resulting numerical values were subsequently normalized.

Book title and Author: Textual features were processed using Term Frequency-Inverse Document Frequency (TF-IDF) to quantify the importance of words in the context of the dataset.

Year-Of-Publication: We calculated the difference between the current year (2024) and the publication year and normalized the resulting values.

There are 49631 data and 3143 features, and Figure 2 called "Overview of data" reveal the information of data after data preprocessing.

2.3 Model description

According to Wei et al. (2012), Logistic regression is a type of regression that is specifically applied when the outcome variable is binary (indicating the presence or absence of an event, and the predictor variables can be either continuous, categorical, or a combination of both [6]. The model calculates the log-odds of the outcome as a linear combination of the input features. This model is simple and interpretable, making it easy to understand the influence of each feature on the prediction.

Decision tree is a model that splits data into subsets based on feature values, making decisions at each node until a prediction is reached at the leaf nodes. It uses criteria like Gini impurity or information gain to determine the best splits. decision trees are intuitive and easy to visualize, but they can overfit, especially on small datasets.

As cited by Lei and Liu (2022), The random forest method is known for its strong generalization capabilities and simplicity. It has been widely applied to tackle complex problems across various tasks and fields [7]. Each tree is trained on a random sample of the data, and predictions are made by averaging the outcomes of individual trees. This approach is effective for large datasets with high dimensionality and noisy features, offering better generalization than a single decision tree.

Table 2: Parameters of deep crossing model.

Layer	Units	Activation	Dropout rate
First hidden	512	ReLU	0.3
Second hidden	256	ReLU	0.3
Third hidden	128	ReLU	0.3
Output Layer	1	ReLU	

The deep crossing model is designed to effectively handle high-dimensional, sparse data, which is typical in recommendation systems. Figure 2 called "Deep crossing model structure illustrates the detailed structure deep crossing model and Table 2 called "Parameters of deep crossing model" shows the parameters for constructing the model.

Hybrid models combine multiple base models to enhance performance by leveraging their strengths and addressing individual weaknesses. By integrating predictions from different models, they improve generalization and predictive accuracy.

These models often use techniques like weighted averaging, stacking, or voting to merge outputs. They are effective on large datasets as they balance bias and variance, leading to improved overall performance.

Here we use weighted averaging method and stacking for comparison, because cascading method did not significantly enhance the final model accuracy. Based on yuan et al. (2021), the essence of recommendation using weighted classification depends on uncovering the explicit or implicit connections between items [8].

The stacking method with cross-validation generates meta-features for both the training and test sets. It splits the training data into n_folds using KFold cross-validation. For each fold, the base model is trained on a subset of the training data, and predictions are made for both the remaining validation data and the test data. The predictions for the validation set are stored as part of the second-level training data, while the predictions for the test set are accumulated across folds. After all folds are completed, the final test predictions are averaged across folds to create robust predictions. Both the second-level training predictions (from the validation sets) and the averaged test predictions are used as meta-features for training the second-level (meta) model in the stacking ensemble.

Table 3: Single model comparison.

Model selection	Accuracy (%)	Variance
Logistic regression	73.43	0.0000795
Decision tree	68.70	0.0003346
Random forest	75.08	0.0002358
Deep crossing	74.73	0.0000890

This technique enhances the model's generalization by combining the predictive power of multiple base models.

2.4 Improve model accuracy and robustness

To address the issue of sample imbalance in the dataset, class weights were applied to the model. This technique ensures that the model assigns greater importance to the underrepresented classes, thereby helping to mitigate the bias towards the majority classes during training.

In addition to using class weights, the dataset was divided into training, validation, and test sets using five different random seeds to create multiple splits. This approach ensures that the model is evaluated on a variety of data partitions, reducing the risk of overfitting to a specific split and providing a more comprehensive assessment of its generalization ability. The entire experimental model was run using a cross-validation method to ensure the results were consistent and that the model's performance was evaluated fairly across all data.

After training the model on these different splits, the average accuracy across the five random seed configurations will be calculated, along with the variance in accuracy. This process provides a more robust measure of the model's performance.

3 Result and Discussion

The tables below illustrate the comparative performance of various individual and hybrid models based on their accuracy. The models included in this comparison are logistic regression, decision tree, deep crossing, and random forest, both individually and in various combinations. The hybrid models are weighted combinations of these base models, designed to leverage the strengths of each to improve overall accuracy. The tables clearly show how different weighting strategies and model combinations impact the accuracy, highlighting that some hybrid approaches significantly outperform individual models, especially when combining deep crossing with other methods. Notably, the highest accuracy is observed with a specific combination of deep crossing, logistic regression, and random forest, demonstrating the effectiveness of this ensemble approach.

3.1 Comparison for singer model

Table 3 named "Single model comparison" presents the accuracy of four individual models: decision tree (DT), logistic regression (LR), deep crossing (DC), and random forest (RF). Among these, the Random Forest model achieves the highest accuracy at 75.08%, while the decision tree model has the lowest accuracy at 68.7%.

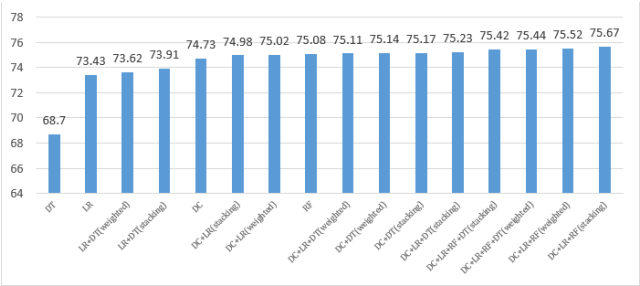

Figure 3: Model accuracy comparison.

3.2 Comparison for two models' combination

Table 4 called "Two models combination" compares the performance of three hybrid model combinations. The models used are logistic regression (LR), decision tree (DT), and deep crossing (DC). The first combination, LR (0.5) + DT (0.5), has a weighted accuracy of 73.62% and a stacking accuracy of 73.91%, with variances of 0.0002813 and 0.0002637 for weighted and stacking accuracy, respectively. The second combination, DC (0.9) + LR (0.1), achieves a slightly higher accuracy at 75.00% with a stacking accuracy of 74.98%, and has lower variance, indicating better consistency. The third combination, DC (0.8) + DT (0.2), demonstrates the highest accuracy values with 75.14% for weighted accuracy and 74.85% for stacking accuracy.

3.3 Comparison for three or more models' combination

Table 5 titled "Three or more models combination "explores the performance of more complex model combinations, involving three or more models. The combination DC (0.7) + LR (0.2) + DT (0.1) shows a weighted accuracy of 75.11% and a stacking accuracy of 75.23%, with variances of 0.0003128 and 0.0002943, respectively. The second combination, DC (0.5) + LR (0.2) + RF (0.1) + DT (0.2), includes random forest (RF) and achieves slightly higher results with a stacking accuracy of 75.42%. The third and most complex combination, DC (0.7) + LR (0.1) + RF (0.2), achieves the highest accuracy of 75.67% for stacking, with variances indicating relative stability compared to other combinations.

3.4 Discussion

From Figure 3 named "Model accuracy comparison", the highest accuracy was achieved with the following combination with stacking: Deep Weight: 0.70, Lr Weight: 0.10, Rf Weight: 0.20, Accuracy: 0.7567.

The first reason is that the deep learning model excels in capturing complex nonlinear relationships and feature interactions. In this combination, the deep model has a relatively high weight (0.7), which indicates that it played a major role in the overall prediction. The deep model's powerful ability allows it to handle complex user behavior patterns and product features, leading to improved prediction accuracy.

Secondly, although the logistic regression model has a lower weight (0.1), it offers good interpretability and robustness. Logistic regression is adept at handling linear relationships and performs

Table 4: Two models combination.

Hybrid model selection with weight	Weighted Accuracy (%)	Variance	Stacking Accuracy (%)	Variance
LR (0.5) + D*T (0.5)*	73.62	0.0002813	73.91	0.0002637
DC (0.9) + LR (0.1)	75.02	0.0001894	74.98	0.0001774
DC (0.8) + DT (0.2)	75.14	0.0003173	75.17	0.0003026

Note: LR means logistic regression, DT means decision tree, DC means deep crossing

Table 5: Three or more models combination.

Hybrid model selection with weight	Weighted Accuracy (%)	Variance	Stacking Accuracy (%)	Variance
DC (0.7) + LR (0.2) + DT (0.1)	75.11	0.0003128	75.23	0.0002943
DC (0.5) + LR (0.2) + RF (0.1) + DT (0.2)	75.44	0.0002996	75.42	0.0002879
DC (0.7) + LR (0.1) + RF (0.2)	75.52	0.0003056	75.67	0.0003041

Note: LR means logistic regression, DT means decision tree, DC means deep crossing, RF means random forest

well when dealing with noisy data, providing a stable foundation for the overall model, thereby avoiding overfitting.

Moreover, the random forest model excels in handling high-dimensional data and avoiding overfitting. In this combination, the random forest's weight is 0.2, meaning it successfully captured the diversity and complexity in the data through its ensemble of multiple decision trees, providing important supplementary decisions to the overall model. This ensemble method effectively reduces bias that might arise from individual models, improving overall prediction performance.

Lastly, stacking utilizes a meta-learner to optimize the outputs of the base models, whereas the weighted method directly applies predefined weights to the models' predictions. As a result, stacking offers more flexibility in dynamically adjusting to complex patterns in the data, which can improve overall performance. In contrast, the fixed weights in the weighted model limit its ability to adapt to uncertainties and non-linear relationships in the outputs. It will lead to slightly lower accuracy compared to the stacking approach.

3.5 Drawbacks and improvements

For the combination model with Deep Learning, logistic regression, and random forest weights set to 0.70, 0.10, and 0.20 respectively, while the overall performance is good, there are some potential drawbacks:

Firstly, combining multiple models with a significant weight given to a deep learning model may increases the overall complexity, which can lead to longer training times and higher computational costs. Thus, it will become less efficient for real-time or large-scale applications.

In addition, although the random forest model has ability to avoid overfitting, the inclusion of a heavily weighted deep learning model (which is prone to overfitting, especially with complex datasets) can still overfit. This could result in the model performing well on the training data but not generalizing as effectively to unseen data.

4 Conclusion

This study conducted a comprehensive analysis of various recommendation models, focusing on the Amazon platform dataset. The models examined include logistic regression, decision tree, random forest, and deep crossing, along with their hybrid combinations. Through careful experimentation and weighted combinations of these models, the deep model combined with logistic regression and random forest yielded the highest accuracy of 0.7567.

The deep learning model's ability to capture complex nonlinear relationships, combined with logistic regression's robustness and random forest's ensemble approach, effectively leveraged the strengths of each method. The study demonstrates how different models complement each other, which may result in improved predictive performance. By balancing complexity and robustness, the hybrid model achieved superior accuracy so that it will be suitable for real-world applications in e-commerce recommendation systems. The findings underscore the importance of model selection and the benefits of hybrid approaches in optimizing recommendation system performance.

References

[1] Latha, Y. M., & Rao, B. S. 2024. Amazon product recommendation system based on a modified convolutional neural network. *ETRI Journal, 46*(4), 633–647. https://doi.org/10.4218/etrij.2023-0162

[2] Wang, Y., Feng, D., Li, D., Chen, X., Zhao, Y., & Niu, X. 2016. A mobile recommendation system based on logistic regression and Gradient Boosting Decision Trees. 2016 International Joint Conference on Neural Networks (IJCNN), 2016-, 1896–1902. https://doi.org/10.1109/IJCNN.2016.7727431

[3] Wu, Y. 2021. The Evolution of Chinese Version Amazon-Taobao's Recommendation System Attention Mechanism. ProQuest Dissertations & Theses,.

[4] Hu, C. 2017. Application of Neural Network Based Recommendation System. ProQuest Dissertations & Theses,.

[5] Yun, Y., Hooshyar, D., Jo, J., & Lim, H. 2018. Developing a hybrid collaborative filtering recommendation system with opinion mining on purchase review. Journal of Information Science, 44(3), 331–344. https://doi.org/10.1177/0165551517692955

[6] Wei, D., Wang, T. & Wang, J. A logistic regression model for Semantic Web service matchmaking. Sci. China Inf. Sci. 55, 1715–1720, 2012. https://doi.org/10.1007/s11432-012-4591-x

[7] Lei, J., & Liu, Q. 2022. Self-paced learning-assisted regularization reconstruction method with data-adaptive prior for electrical capacitance tomography. Expert Systems with Applications, 191, 116296-. https://doi.org/10.1016/j.eswa.2021.116296

[8] Yuan, Z., Lee, J. H., Zhang, S. 2021. Optimization of the Hybrid Movie Recommendation System Based on Weighted Classification and User Collaborative Filtering Algorithm. Complexity (New York, N.Y.), 2021(1). https://doi.org/10.1155/2021/4476560